



HK IMO Training 2021-2022
Problem Set 20
For 5 March 2022



Problem 77. Let n be a positive integer. Let $\{x\} = x - \lfloor x \rfloor$ be the fractional part of a positive real number x . Prove that

$$\left| \{\sqrt{n}\} - \frac{1}{2} \right| > \frac{1}{8(\sqrt{n} + 1)}.$$

Problem 78. Let $\mathbb{Z}^+$ be the set of all positive integers. Find all function(s) $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ such that $m \mid n$ if and only if $f(m) \mid f(n) - m$ for any positive integers m and n .

Problem 79. Let $n > 1$ be an integer. Two players X and Y play the following game on an $n \times 2021$ chessboard. Firstly, X colours every cell black or white. Secondly, Y places a chess piece in one of the cells in the top row, and designates one of the cells in the bottom row as the terminal cell. Then X moves the chess piece in the following way:

- If the chess piece lies in a white cell, X has to move it to the cell below.
- If the chess piece lies in a black cell, X can move it to the left cell or the right cell (if the cell exists), and then move it to the cell below.

Find the smallest possible value of n such that X can always move the chess piece to the terminal cell.

Problem 80. Let $\triangle ABC$ be an acute triangle with circumcentre O . The A -symmedian of $\triangle ABC$ meets the circle with diameter BC at the points D and E (where D lies closer to A than E), and meets the circumcircle of $\triangle ABC$ again at F . Let P be the intersection point of BC and FO , and let Q be the intersection of the line EP and the median AM of $\triangle ABC$. Prove that AP bisects the segment DQ .